SUHAS NAGARAJ

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EDUCATION

Master of Engineering in Robotics [Minor: Artificial Intelligence] [CGPA: 4.0 / 4.0] — **August 2023 - May 2025**
University Of Maryland, College Park [Phi Kappa Phi Honor Society Member] — College Park, United States

Bachelor of Engineering in Mechanical Engineering [CGPA: 9.3 / 10.0] — **August 2017 - July 2021**
Ramaiah Institute of Technology — Bengaluru, India

PROFESSIONAL EXPERIENCE

Research Intern (On-site + Remote) — **May 2024 – September 2024**
Intelligent Inclusive Interaction Design (I3D) Lab, Indian Institute of Science (IISc) — Bengaluru, India

- Conducted Research & Development work under Dr. Pradipta Biswas (Robert Bosch Centre for Cyber Physical Systems)
- Developed a novel 3D reconstruction algorithm integrating LiDAR sensors with 3D Gaussian Splatting SLAM and ROS2, achieving a 200% improvement in localization accuracy (measured as RMSE) and a 35% reduction in computational load, significantly enhancing autonomous navigation and mapping capabilities; Worked with Turtlebot3 Waffle, Tortoisebot and Tankbot mobile robots for integration

Engineering Intern — **October 2022 – March 2023**
DiFACTO Robotics and Automation — Bengaluru, India

- Excelled in designing automation parts using SolidWorks reducing design time by 5%; conducted industrial process simulations with FANUC RoboGuide leading to a 25% increase in operational efficiency; Gained proficiency in offline and online programming of industrial robots (ABB RAPID and FANUC KAREL); Contributed to the assembly of Automated Guided Vehicles deployed in the Maruti Suzuki assembly line.
- Developed skills in PLC programming and Devicenet interfacing to effectively control robot cells using Mitsubishi PLC.

Technical Consultant — **August 2021 – March 2022**
U-Solar Clean Energy Solutions Pvt. Ltd. — Bengaluru, India

- Managed more than 50 client and 10 partner accounts, accelerated sales as part of the inaugural team of Technical Consultants, contributing to a 20% increase in customer retention; Spearheaded market and policy research initiatives, resulting in a 15% growth in market reach. Devised and implemented strategies that drove a 25% increase in leads and projects in the pipeline

SKILLS

- **Technical Expertise:** ROS1/ROS2, Computer Vision, Machine Learning, Neural Networks [Feedforward, CNN, Sequence and Transformers], Natural Language Processing, SLAM, 3D mapping and reconstruction, Data Analytics, Data Structures and Algorithms, Hardware, Sensor Fusion, Multi-threading, Optimization, Control theory, Design, Rapid Prototyping, Could Computing, Containerization, Version Control
- **Programming Languages:** Python, C++, MATLAB, RAPID (ABB), KAREL (FANUC), PLC Programming.
- **Libraries, Frameworks and Tools:** PyTorch, Cuda, Tensorflow, Keras, ONNX, Huggingface, OpenCV, Open3D, PCL, Albumentations, Yolo, Cartographer, NumPy, pandas, scikit-learn, NLTK, SQL, Tableau, Seaborn, Kubernetes, Docker, AWS, Linux, GitHub.
- **Others:** MATLAB Simulink, FANUC ROBOGUIDE, ABB Robot Studio, CimStation Robotics, SolidWorks, AutoCAD, Fusion 360, CATIA.

PROJECTS [www.suhasnagaraj.com/projects]

- **Adaptive Text-to-Command Translation for Robotic Navigation:** Fine-tuned T5-Small and LoRA models to translate natural language commands into structured navigation plans for TurtleBot3 in ROS2 and Gazebo
- **ASD Prediction Using Eye Tracking Data:** Developed a novel deep learning approach (CNN, LSTM, Transformers) for classifying Autism Spectrum Disorder using eye-tracking data, achieving 88.9% accuracy.
- **Automated Guided Vehicle:** Designed autonomous navigation with lane following, stop sign detection, and dynamic obstacle avoidance for TurtleBot3 using ROS2 and OpenCV in simulation and real-world setups.
- **Maze Navigation with Object Detection:** Developed ROS2-based system for maze traversal using Aruco markers and object detection
- **SLAM-Based Waypoint Navigation:** Designed an autonomous waypoint navigation system for TurtleBot3 using SLAM, TF tree localization.
- **Deep Reinforcement Learning:** Applied Deep Q-Networks (DQN) for obstacle avoidance and autonomous navigation in a Turtlebot3.
- **Path Planning Algorithms:** Developed Dijkstra's, A*, RRT, RRT*, Bidirectional A*, Potential Field and Real Time RRT* algorithms for mobile robot navigation, implemented the solutions on a Turtlebot3 Waffle robot.
- **Computer Vision:** Built object tracking, edge and corner detection, panoramic image stitching, and stereo depth mapping solutions using classical computer vision techniques; Performed camera calibration to analyze reprojection errors and correct image distortion.
- **CNC Tending Bot:** Analyzed and implemented the kinematics of the ABB IRB 1600 robot for CNC machine tending.
- **Control Systems:** Designed and implemented LQR and LQG controllers for a dynamic cart system; Analyzed, implemented and compared Feedback Linearization & Adaptive Sliding Mode control strategies for quadrotor helicopters

PUBLICATION

3D Reconstruction via Camera-Lidar (2D) Fusion for Mobile Robots: A Gaussian Splatting Approach
*Ajay Kumar Sandula, *Shriram Damodaran, ***Suhas Nagaraj**, Debasish Ghose and Pradipta Biswas [* are equally contributing authors]*
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